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A Proofs and lemmas in Section 3

A.1 Lemma on the $\mathcal{W}_2$ -(sub)gradient

Lemma A.1. Suppose $G : \mathcal{P}_2(X) \rightarrow (-\infty, +\infty]$ is lower semi-continuous and $\text{Dom}(G) \subset \mathcal{P}_2^r$. Let $\gamma > 0$, $p \in \mathcal{P}_2^r$, and

$$F(\rho) = G(\rho) + \frac{1}{2\gamma} \mathcal{W}_2^2(p, \rho). \quad (\text{A.1})$$